

Problem Sets Between Test 1 and Test 2

Only turn in problems that are **not** bracketed. Bracketed problems are additional problems you can look at. Round brackets indicate problems that may help you with problems that are assigned; square brackets are additional problems on material that you should know, but you are not required to write up solutions; curly brackets are truly optional and may contain extra nuggets that you will not be required to know but may be interested in.

Additional assignments will be filled in over time.

notation	meaning
unbracketed	assigned problem – turn these in for grading
()	helper/warm-up problem
[]	additional problems (you are responsible for content, but don't turn them in)
{ }	covers optional material

Due	Task	Source	Problems
Mon 3/16	WW06	Rosen 7	Probability
Wed 3/18	WW07	Rosen 7	Probability
Tue 3/24	WW08 PS 7	Rosen 7 Rosen 7.1 Rosen 7.2	Probability 26b lottery 28 super lottery [1-33 odd] [30] lottery [4] April [6] A or ♡ [8] ♡A [18] straight flush [35] roulette 8 permutation events 25 bit strings 28 children 31b children [1] [2] loaded die (5-7) permutation events [19] birth month [20] birth day
Wed 4/1	WW09	Rosen 7	Due at 11 AM (ie, before class time)
Fri 4/3	PS 8 WW10	Rosen 7.2 Rosen 7.3 Rosen 7	29 odd one out (1) (3) (5) 2 4 6 steroids [8] rare disease [9] HIV [13] [14] [15] Monty Hall Expected Value
Mon 4/6	PS 9 WW11	Rosen 7.4 Rosen 7	(1) 3 rolling 6s [5] loaded die 7 test score 8 three dice Expected Value; binomial distributions
Wed 4/8	PS 10	Rosen 13.1	1ad sentences [2] (3) 4 [5] 6a 9 (13) 14bd [15]
Mon 4/13	PS 11	Rosen 13.1	19 types of grammars 20 palindromes 21a union Be sure to provide an explanation for problem 19.
Wed 4/15	PS 12	Rosen 13.3	(11) testing strings 12 testing strings 16-17 what language? 25-26 create DFA
Fri 4/17	PS 13	Rosen 13.3	19-20 what language? [32] create DFA 34 create DFA 36 create NFA 45-46 what language? [47] what language? 55-56 create DFA, NFA Note: Please do 55 and 56 in this order: 55a, 56a, 55b, 56b, 55c, 56c.
Mon 4/20	PS 14	Rosen 13.3 Rosen 13.4	52-54 NFA to DFA [1] 2abde describe in words [3] 4 does it belong? 6 regex [7] regex

Due	Task	Source	Problems
Fri 4/24	PS 15	Rosen 13.4 Rosen 7.SE	10 build NFA 12 Kleene's Thm [13] Kleene's Thm [20] lonely start 6ab cards
Mon 4/27	PS 16	Rosen 13.4 Rosen 7.3 Rosen 7.SE	14abc grammar $\rightarrow$ NFA 15–16 NFA $\rightarrow$ grammar [17] NFA $\rightarrow$ grammar 10 flu test 24 large/small
Review	–	Rosen 6.RQ Rosen 6.SE Rosen 7.RQ Rosen 7.SE Rosen 13.RQ Rosen 13.SE	[1–2, 5, 8] [1–7] [1–6, 11] [1–5, 7a, 17, 23] [1–2] [4] [5a] [6a–12] [7] [8] [19–21] [23–25]